Universidad Internacional del Ecuador (UIDE)

School of Computer Science

Network Communication Technology

Title: *Week 9 Prototype: Initial IPv4 Network for the Two-Site Enterprise Chat Project in Cisco Packet Tracer*

Team members

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Repository: https://github.com/ales231/Integrative_project_

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1. Introduction

1.1 Purpose and Scope of the Prototype

This report documents the initial IPv4 prototype of the two-site network for the enterprise chat project (Go chat server, FastAPI authentication, and a network of two sites). The prototype was built and tested in Cisco Packet Tracer 9.0.1 (file *week9_prototype.pkt*). Because VLANs, trunks, and EtherChannel have not yet been implemented by the team, the prototype covers only the **active segments**: the user LAN of each site, the shared access-point segment between the sites, DHCP for the two user LANs, and temporary static routes between the sites. Everything else in the Phase 1 design is marked **PENDING** and none of it is reported as passed.

In the simulation, each site router is a Cisco 2911 that stands in for the OpenWrt router (Raspberry Pi) of the final design, and a Cisco 2960 switch (SW-AP) represents the shared access-point segment 192.168.100.0/24. The access point itself (.1) is not modelled. The source code and documentation are in the project repository: https://github.com/ales231/Integrative_project_. No passwords, tokens, or keys are included in this report or in the delivered files.

1.2 Updated Architecture

The architecture of Phase 1 is kept: Site A (HQ Quito, 10.10.0.0/22) and Site B (branch Guayaquil, 10.10.4.0/22), one router per site, and a shared segment between the routers. Figure 1 shows what is implemented (solid green) and what is pending (dashed gray), with the interface labels. Table 1 compares the plan with the prototype and gives the reason for each change.

Table 1

Planned Elements Versus the Week 9 Prototype

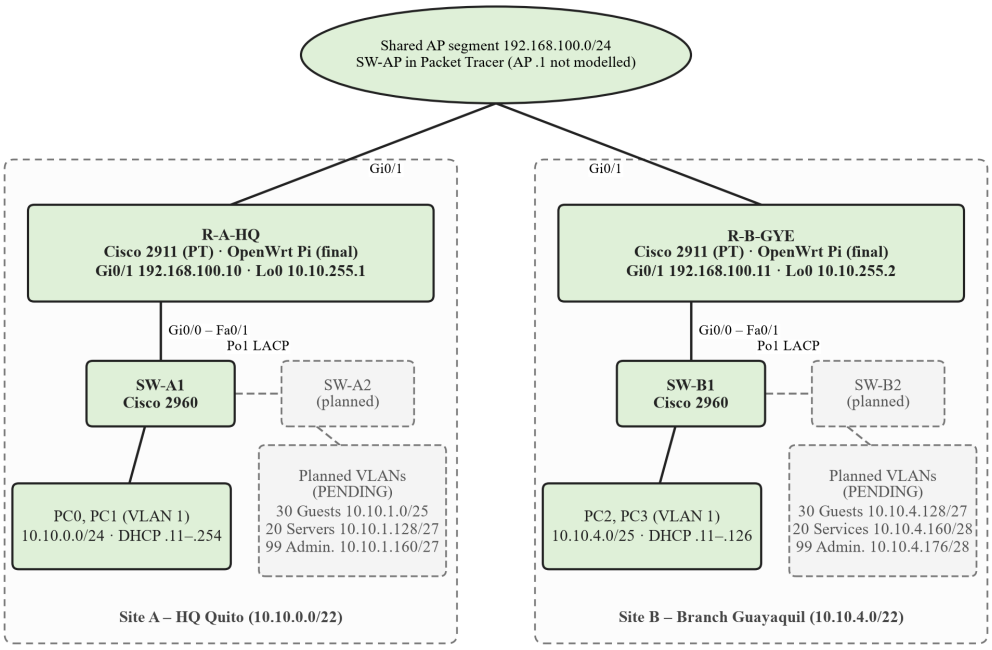
Element	Phase 1 plan	Week 9 prototype	Status	Reason for the change
Sites and blocks	10.10.0.0/22 and 10.10.4.0/22	Same blocks; only the user subnets are in use	Partial	Segmentation is pending
Site routers	OpenWrt on Raspberry Pi	Cisco 2911 (R-A-HQ, R-B-GYE)	Simulated	Packet Tracer has no OpenWrt device
Shared segment	192.168.100.0/24, AP .1, routers .10 and .11	SW-AP switch, routers .10 and .11; AP not modelled	Implemented	The AP only forwards frames
Loopbacks	10.10.255.1 and .2	Configured on both routers	Implemented	—
Switches per site	Two (root and backup)	One per site (SW-A1, SW-B1)	Partial	EtherChannel and RSTP are pending
VLANs 10/20/30/99, native 999	Four VLANs per site	All ports in VLAN 1	PENDING	VLAN content not yet implemented
Trunk and Router-on-a-Stick	eth0.N subinterfaces	Router port Gi0/0 is a plain routed port	PENDING	Depends on VLANs
Po1 (LACP), RSTP	4 × 1 Gbps, root at SW-x1	Not configured	PENDING	Needs the second switch
Inter-site routing	OSPFv2 area 0	Static routes (temporary)	Changed	OSPF is not used in this delivery; see Section 2.4
DHCP	User and guest networks	User networks of both sites	Partial	Guest networks need VLAN 30
NAT and firewall	Per-site NAT, fw4 rules	Not configured	PENDING	No Internet segment in the prototype
Secure administration	Hashed credentials, SSH	Hashed secrets and local login on all five devices; SSH on both routers	Partial	Switch SSH needs a management IP (VLAN 99)

Note. PENDING = not implemented and not tested in this delivery.

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Figure 1

Updated Architecture: Implemented and Pending Elements



Green solid = implemented and tested in Packet Tracer (week 9). Gray dashed = planned, status PENDING.

Note. Interface names are those of the Packet Tracer devices. Po1 = port channel planned between the two switches of each site.

2. Proposed Methodology

2.1 Addressing Plan (VLSM from 10.10.0.0/16)

The plan was derived from the number of devices per segment plus 25% growth and one gateway address, allocating the largest subnet first inside each site block. The calculation is reproducible with *plan_data.py*, which also checks that no subnets overlap. Table 2 lists the eight planned subnets; distinct subnets are reserved for the VLANs that are still pending. Addresses from 10.10.1.192 (Site A) and 10.10.4.192 (Site B) remain free.

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Table 2

VLSM Addressing Plan for Both Sites

Site	VLAN	Purpose	Network / prefix	Mask	Usable range	Broadcast	Gateway	Assignment
HQ	10	Users	10.10.0.0/24	255.255.255.0	10.10.0.1–254	10.10.0.255	10.10.0.1	DHCP, pool 11–254
HQ	30	Guests	10.10.1.0/25	255.255.255.128	10.10.1.1–126	10.10.1.127	10.10.1.1	DHCP, pool 11–126
HQ	20	Servers (Go, FastAPI, DB)	10.10.1.128/27	255.255.255.224	10.10.1.129–158	10.10.1.159	10.10.1.129	Static
HQ	99	Administration	10.10.1.160/27	255.255.255.224	10.10.1.161–190	10.10.1.191	10.10.1.161	Static
BR	10	Users	10.10.4.0/25	255.255.255.128	10.10.4.1–126	10.10.4.127	10.10.4.1	DHCP, pool 11–126
BR	30	Guests	10.10.4.128/27	255.255.255.224	10.10.4.129–158	10.10.4.159	10.10.4.129	DHCP, pool 139–158
BR	20	Local services	10.10.4.160/28	255.255.255.240	10.10.4.161–174	10.10.4.175	10.10.4.161	Static
BR	99	Administration	10.10.4.176/28	255.255.255.240	10.10.4.177–190	10.10.4.191	10.10.4.177	Static

Note. HQ = Site A (Quito); BR = Site B (Guayaquil). Only the two VLAN 10 user subnets are active in the prototype, carried untagged in VLAN 1. The loopbacks are 10.10.255.1 (R-A-HQ) and 10.10.255.2 (R-B-GYE).

2.2 Basic Configuration

The devices were named R-A-HQ, R-B-GYE, SW-A1, SW-B1, and SW-AP. Each router received an address on its user LAN (Gi0/0), an address on the shared segment (Gi0/1), and a loopback. The switches keep their default VLAN 1 and carry only descriptions on the used ports. Table 3 lists the active interfaces; the complete configurations are delivered as text files (one per device).

Table 3

Active Router Interfaces in the Prototype

Device	Interface	Address	Purpose	Connected to
R-A-HQ	Gi0/0	10.10.0.1/24	Users LAN gateway (Site A)	SW-A1 Fa0/1
R-A-HQ	Gi0/1	192.168.100.10/24	Shared AP segment	SW-AP Fa0/1
R-A-HQ	Loopback0	10.10.255.1/32	Router identifier	—
R-B-GYE	Gi0/0	10.10.4.1/25	Users LAN gateway (Site B)	SW-B1 Fa0/1
R-B-GYE	Gi0/1	192.168.100.11/24	Shared AP segment	SW-AP Fa0/2
R-B-GYE	Loopback0	10.10.255.2/32	Router identifier	—

2.3 DHCP

Each router is the DHCP server of its user LAN. The first ten addresses of each subnet are excluded (gateway and static devices), and the pool starts at .11, as in the plan. Pools for guests (VLAN 30) are pending because they need their own VLAN.

The pools are named HQ-USUARIOS and GYE-USUARIOS (Table 5); the complete definitions are in the delivered configuration files.

2.4 Inter-Site Routing (Temporary Static Routes)

The two sites reach each other through the shared segment with static routes, chosen for this delivery instead of OSPF. They only cover the active subnets and the loopbacks (Table 4). They are temporary: in Phase 3 they will be replaced by OSPFv2 area 0 on the shared segment (FRRouting on the OpenWrt routers; FRRouting, n.d.), which will advertise every VLAN subnet automatically (Moy, 1998). Until then, each new VLAN would need its own static route.

Table 4

Static Routes Configured

Router	Destination	Next hop	Purpose
R-A-HQ	10.10.4.0/25	192.168.100.11	Users LAN of Site B
R-A-HQ	10.10.255.2/32	192.168.100.11	Loopback of R-B-GYE
R-B-GYE	10.10.0.0/24	192.168.100.10	Users LAN of Site A
R-B-GYE	10.10.255.1/32	192.168.100.10	Loopback of R-A-HQ

2.5 Secure Administration

All five devices have a banner, *no ip domain-lookup*, the domain *proyecto.local*, an RSA key of 2048 bits, and SSH version 2. Privileged mode is protected with *enable secret* and a local user *admin* is stored with *secret*, so only salted hashes appear in the configuration (no *enable password* and no *username ... password* line exists). The console and all VTY lines use *login local*, the VTY lines accept only SSH, and idle sessions end after five minutes. On the switches, every unused port is shut down. *service password-encryption* is also active, but it is reversible and adds nothing now that the secrets are hashed.

PENDING: SSH to the switches cannot be tested until they receive a management address in VLAN 99 (week 10). Also pending are an access list on the VTY lines and replacing the current secrets with stronger, unique ones for each device before submission (a team action). The secrets were typed by the team directly on the devices; they appear in no file, and the configuration copies for the repository have the hashes replaced with a placeholder.

3. Results / Development

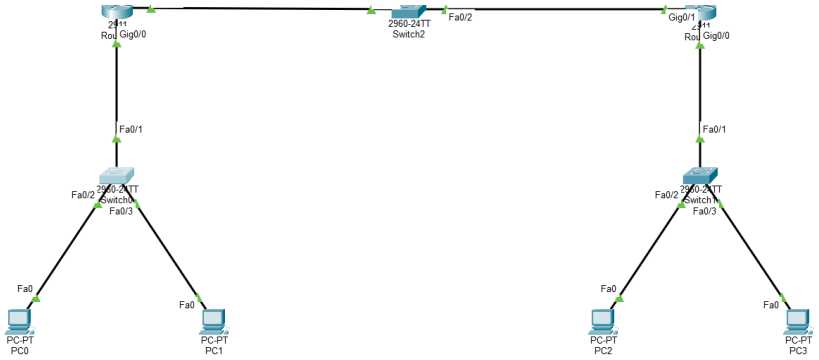
3.1 Packet Tracer Topology

Figure 2 shows the topology built in Packet Tracer with the port labels. All links came up (green) after the interfaces were configured.

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Figure 2

Packet Tracer Topology With Port Labels



3.2 Interface Status and MAC Address Tables

Fa0/1 to Fa0/3 of SW-A1 and SW-B1 are *connected* in VLAN 1 (full duplex, 100 Mbps), and the other ports are *notconnect*. The MAC tables show how each switch forwards: the address learned on Fa0/1 belongs to the device behind the uplink (the router's Gi0/0), and the addresses on Fa0/2 and Fa0/3 belong to the PCs. For example, 000c.8589.b8b1 and 000c.8503.7303 on SW-A1 match the DHCP bindings of 10.10.0.11 (PC0) and 10.10.0.12 (PC1) on R-A-HQ, and 000a.410a.b019 on SW-B1 matches the binding of 10.10.4.12 (PC3). The entries are dynamic, so a PC appears only after it has sent a frame and disappears when it ages out: at the time of the capture SW-B1 listed no entry for PC2 (0060.2f10.8a10, bound to 10.10.4.11). On SW-AP, Fa0/1 and Fa0/2 hold the MAC addresses of the two routers' Gi0/1 interfaces (identified from the topology and the consecutive router MAC addresses, since the router MACs were not printed).

Figure 3

SW-A1: Interface Status and MAC Address Table

```
SW-A1#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1     Uplink a R-A-HQ Gi connected    1          a-full  a-100 10/100BaseTX
Fa0/2     PC-A1 usuarios     connected    1          a-full  a-100 10/100BaseTX
Fa0/3     PC-A2 usuarios     connected    1          a-full  a-100 10/100BaseTX

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.8503.7303    DYNAMIC     Fa0/3
   1    000c.8589.b8b1    DYNAMIC     Fa0/2
   1    0090.0c05.7d01    DYNAMIC     Fa0/1
SW-A1#
```


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Figure 4

SW-B1: Interface Status and MAC Address Table

```
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1     Uplink a R-B-GYE G connected    1          a-full  a-100 10/100BaseTX
Fa0/2     PC-B1 usuarios     connected    1          a-full  a-100 10/100BaseTX
Fa0/3     PC-B2 usuarios     connected    1          a-full  a-100 10/100BaseTX
SW-B1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0007.ecb1.8401    DYNAMIC     Fa0/1
   1    000a.410a.b019    DYNAMIC     Fa0/3
SW-B1#
```

Figure 5

SW-AP: MAC Address Table

```
SW-AP>
SW-AP>show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0007.ecb1.8402    DYNAMIC     Fa0/2
   1    0090.0c05.7d02    DYNAMIC     Fa0/1
SW-AP>
```

3.3 DHCP Results

All four PCs obtained an address with the message *DHCP request successful* (Figure 6). The bindings on the routers match (Figures 7 and 8). The addresses were assigned from .11, so the excluded range was respected.

Figure 6

Addresses Received by the Four PCs (IP Configuration Window)

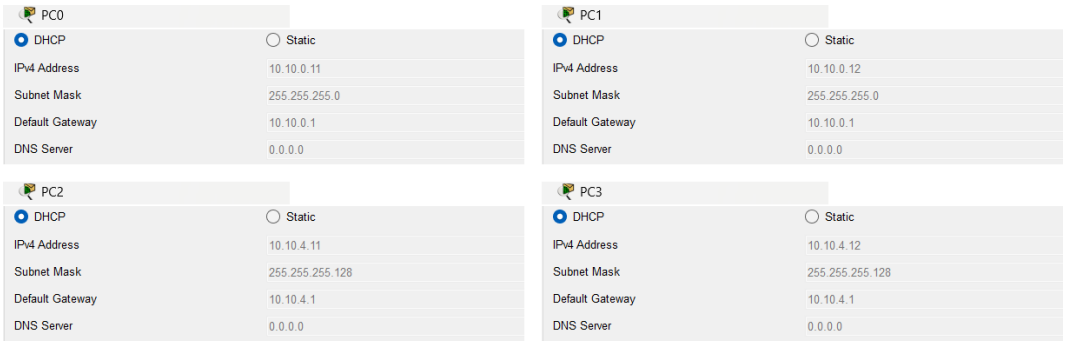


Table 5

DHCP Pools and Clients Observed

Pool	Network	Gateway	Excluded	Client	Address received	MAC (binding)
HQ-USUARIOS	10.10.0.0/24	10.10.0.1	10.10.0.1–.10	PC0	10.10.0.11	000C.8589.B8B1
HQ-USUARIOS	10.10.0.0/24	10.10.0.1	10.10.0.1–.10	PC1	10.10.0.12	000C.8503.7303
GYE-USUARIOS	10.10.4.0/25	10.10.4.1	10.10.4.1–.10	PC2	10.10.4.11	0060.2F10.8A10
GYE-USUARIOS	10.10.4.0/25	10.10.4.1	10.10.4.1–.10	PC3	10.10.4.12	000A.410A.B019

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3.4 Routing Tables and Packet Paths

Both routers show their connected networks (C), local addresses (L), and two static routes (S) learned from the configuration (Figures 7 and 8). The traceroute from PC3 (Site B) to PC0 (Site A) crosses R-B-GYE (10.10.4.1) and R-A-HQ through its shared-segment address (192.168.100.10) before reaching 10.10.0.11 (Figure 9), which is the path the static routes define. The TTL of 126 in the replies is consistent with two routers on the path.

Figure 7

R-A-HQ: Interface Summary, Routing Table, and DHCP Bindings

```
R-A-HQ#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.0.1       YES manual up                    up
GigabitEthernet0/1     192.168.100.10  YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Loopback0              10.10.255.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R-A-HQ#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
C       10.10.0.0/24 is directly connected, GigabitEthernet0/0
L       10.10.0.1/32 is directly connected, GigabitEthernet0/0
S       10.10.4.0/25 [1/0] via 192.168.100.11
C       10.10.255.1/32 is directly connected, Loopback0
S       10.10.255.2/32 [1/0] via 192.168.100.11
     192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.100.0/24 is directly connected, GigabitEthernet0/1
L       192.168.100.10/32 is directly connected, GigabitEthernet0/1

R-A-HQ#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.10.0.11       000C.8589.B8B1           --                     Automatic
10.10.0.12       000C.8503.7303           --                     Automatic
R-A-HQ#
```


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Figure 8

R-B-GYE: Interface Summary, Routing Table, and DHCP Bindings

```
R-B-GYE#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.4.1       YES manual up                    up
GigabitEthernet0/1     192.168.100.11  YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Loopback0              10.10.255.2     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R-B-GYE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S       10.10.0.0/24 [1/0] via 192.168.100.10
C       10.10.4.0/25 is directly connected, GigabitEthernet0/0
L       10.10.4.1/32 is directly connected, GigabitEthernet0/0
S       10.10.255.1/32 [1/0] via 192.168.100.10
C       10.10.255.2/32 is directly connected, Loopback0
     192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.100.0/24 is directly connected, GigabitEthernet0/1
L       192.168.100.11/32 is directly connected, GigabitEthernet0/1

R-B-GYE#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.10.4.11       0060.2F10.8A10           --                     Automatic
10.10.4.12       000A.410A.B019           --                     Automatic
R-B-GYE#
```

Figure 9

PC3: Ping and Tracert to PC0 (First Run With 50% Loss, Then a Clean Run)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.0.11

Pinging 10.10.0.11 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.10.0.11: bytes=32 time<1ms TTL=126
Reply from 10.10.0.11: bytes=32 time<1ms TTL=126

Ping statistics for 10.10.0.11:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>tracert 10.10.0.11

Tracing route to 10.10.0.11 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.10.4.1
  2    0 ms      0 ms      0 ms      192.168.100.10
  3    0 ms      0 ms      0 ms      10.10.0.11

Trace complete.
```

```
C:\>ping 10.10.0.11

Pinging 10.10.0.11 with 32 bytes of data:

Reply from 10.10.0.11: bytes=32 time<1ms TTL=126
Reply from 10.10.0.11: bytes=32 time<1ms TTL=126
Reply from 10.10.0.11: bytes=32 time<1ms TTL=126
Reply from 10.10.0.11: bytes=32 time<1ms TTL=126

Ping statistics for 10.10.0.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

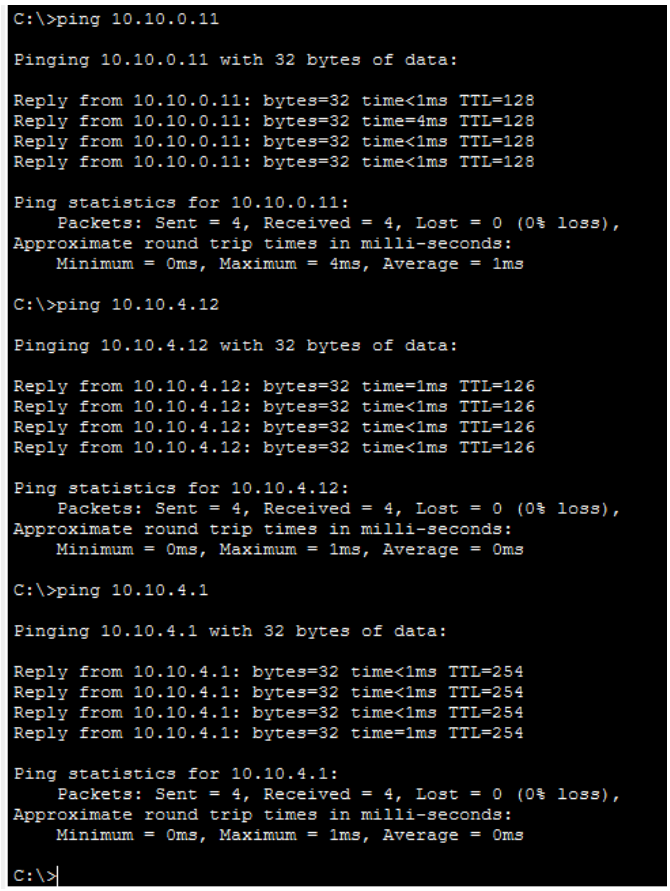
Note. Two regions of the same terminal session; the omitted lines are a repeated ping and tracert with the same result.

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Figure 10

PC1: Ping to PC0, to PC3, and to the Gateway of Site B



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Figure 11

Console Login Tests on the Five Devices



Note. Row by row: SW-A1 and SW-B1; SW-AP and R-A-HQ; R-B-GYE. The captures were cropped so that no secret typed during configuration is visible.

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Figure 12

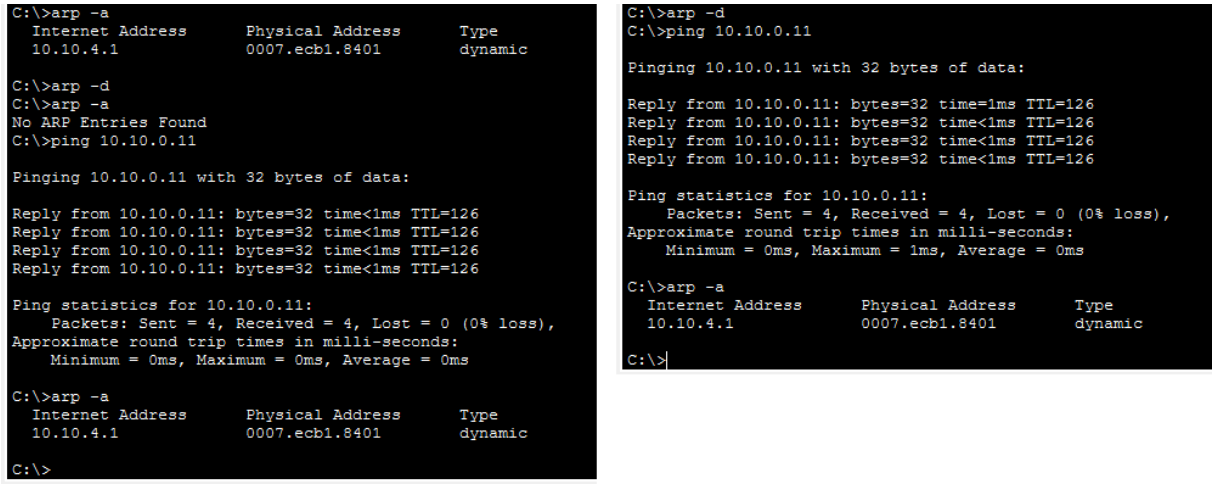
SSH Login Tests From the PCs: R-A-HQ (PC0) and R-B-GYE (PC3)



Note. Left: PC0 to 10.10.0.1 (accepted, then rejected). Right: PC3 to 10.10.4.1 (rejected, then accepted).

Figure 13

PC3: Attempts to Reproduce the First-Ping Loss After Clearing ARP Caches



Note. Left: arp -d on PC3, then ping to PC0 (4/4 replies, 0% loss). Right: a second run after the ARP cache of R-A-HQ was also cleared (4/4 replies, 0% loss). The clear on R-A-HQ was seen on screen but its own capture was not archived.

3.5 Preliminary Test Matrix

Table 6 lists the 14 tests that were executed. The status is *Passed* only when the observed result matched the expected one in the captures. Table 7 lists what could not be tested; none of it is counted as passed.

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Table 6

Executed Tests

ID	Source	Destination	Procedure	Expected	Observed	Evidence	Status
T01	PC1 (10.10.0.12)	PC0 (10.10.0.11)	ping 10.10.0.11	Replies, 0% loss	4/4 replies, TTL 128, 0% loss	Fig. 10	Passed
T02	PC3 (10.10.4.12)	PC0 (10.10.0.11)	ping 10.10.0.11	Replies, 0% loss	First run 2/4 (50% loss: first two echoes timed out); every later run 4/4, TTL 126	Fig. 9, 13	Passed (see P-01)
T03	PC1	PC3 (10.10.4.12)	ping 10.10.4.12	Replies, 0% loss	4/4 replies, TTL 126, 0% loss	Fig. 10	Passed
T04	PC1	R-B-GYE Gi0/0 (10.10.4.1)	ping 10.10.4.1	Replies, 0% loss	4/4 replies, TTL 254, 0% loss	Fig. 10	Passed
T05	PC3	PC0	tracert 10.10.0.11	Path R-B, R-A, destination	Hops 10.10.4.1, 192.168.100.10, 10.10.0.11	Fig. 9	Passed
T06	PC0–PC3	DHCP servers	IP Configuration, DHCP	Address in pool, correct mask and gateway	10.10.0.11, 10.10.0.12 (/24, gw .1); 10.10.4.11, 10.10.4.12 (/25, gw .1)	Fig. 6	Passed
T07	R-A-HQ, R-B-GYE	Local	show ip interface brief	Gi0/0, Gi0/1, Lo0 up/up	All three up/up on both routers; Gi0/2 administratively down	Fig. 7, 8	Passed
T08	R-A-HQ, R-B-GYE	Local	show ip route	Connected and two static routes	C, L, and two S routes on each router	Fig. 7, 8	Passed
T09	R-A-HQ, R-B-GYE	Local	show ip dhcp binding	Two bindings per router	10.10.0.11/.12 and 10.10.4.11/.12 with the PC MAC addresses	Fig. 7, 8	Passed
T10	SW-A1, SW-B1	Local	show interfaces status	Fa0/1–3 connected	Fa0/1–3 connected, VLAN 1, 100 Mbps full duplex	Fig. 3, 4	Passed
T11	SW-A1, SW-B1, SW-AP	Local	show mac address-table	Learned MACs on the used ports	SW-A1: 3 entries; SW-B1: 2 entries (PC2 aged out); SW-AP: 2 entries	Fig. 3–5	Passed
T12	Console	SW-A1, SW-B1, SW-AP, R-A-HQ, R-B-GYE	Log in with a wrong and then the correct password; enable	Wrong password rejected; correct one reaches privileged mode	All five reached privileged mode. A wrong password was rejected (% Login invalid) on SW-A1, SW-B1, SW-AP, and R-A-HQ; the rejected attempt on R-B-GYE is the SSH test T14	Fig. 11	Passed (see P-04)
T13	PC0	R-A-HQ (10.10.0.1)	ssh -l admin 10.10.0.1	Wrong password rejected; correct one opens R-A-HQ>	Correct password opened R-A-HQ>; a second attempt with a wrong password was rejected (% Login invalid)	Fig. 12	Passed
T14	PC3	R-B-GYE (10.10.4.1)	ssh -l admin 10.10.4.1	Wrong password rejected; correct one opens R-B-GYE>	Rejected with % Login invalid, then accepted	Fig. 12	Passed

Note. Figure numbers refer to the screenshots in this report; the raw captures are in the *evidence* folder.

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Table 7

Pending Tests

ID	Test	Reason it is pending	Planned week
P01	VLAN isolation between users, guests, servers, and administration	VLANs not implemented	10
P02	802.1Q trunk and Router-on-a-Stick (inter-VLAN routing)	Depends on P01	10
P03	DHCP for guest networks (VLAN 30) and exclusion checks	VLAN 30 not implemented	10
P04	EtherChannel Po1 failover (LACP) and RSTP root election	Second switch per site not built	10–11
P05	SSH login to the switches (the router SSH test is T13 and T14); VTY access list	Switches have no management IP until VLAN 99	10
P06	OSPF adjacency and route exchange between the sites	Replaces the static routes	11
P07	NAT and firewall behavior per site, Internet access	No Internet segment in the prototype	11
P08	Reverse trace (PC0 to PC3) and static-route failure behavior	Not executed in this session	10
P09	OpenWrt router functions on Raspberry Pi hardware	Hardware not yet available	11
P10	Replace the current secrets with stronger, unique ones and re-run T12 to T14	Team action	10

Note. All entries have status PENDING.

3.6 Implementation Problems

Table 8 records what actually happened during the build, with the diagnosis, the correction, and the retest. P-04 was the only problem that interrupted the work.

Table 8

Problems Observed During the Build

ID	Problem	Diagnosis	Correction	Retest
P-01	First inter-site ping from PC3 to PC0 lost 2 of 4 packets (50% loss)	Cause not determined. ARP resolution was suspected, but clearing the ARP cache on PC3 and on R-A-HQ and pinging again gave 0% loss (Figure 13; the clear on R-B-GYE was not verified), so the loss was not reproduced	None possible without a reproduction; the test was simply repeated	Later runs: 4/4 replies, 0% loss (Figure 9), and after clearing the ARP caches (Figure 13)
P-02	R-A-HQ rejected *no ip http server* (invalid input)	That command is not available in the simulated IOS image of the 2911	Removed from the scripts; R-B-GYE was configured without it	Exported running configuration of both routers contains no HTTP server line
P-03	Text that is not IOS was pasted into the console of R-A-HQ during setup (tooling error)	The clipboard held earlier text when the paste button was used	Name lookup was aborted; the configuration was then applied cleanly	Exported running configuration of R-A-HQ shows only the intended commands
P-04	Console of SW-A1 rejected every login after *login local* was enabled, so the switch could not be managed	The credentials entered at the prompt did not match the secrets that had been configured (typing differences are the most likely cause: passwords are case sensitive and look-alike characters were involved)	A small text file with *no login* on the console was merged through Config, Running Config, Merge; the old secrets were removed the same way; the team created them again and *login local* was enabled only after they were checked	Login accepted on SW-A1 and, with the same order of steps (secrets, check, *login local*, test, save), on the other four devices (T12 to T14)

3.7 Completion and Migration Plan (Cisco to OpenWrt)

The Cisco commands are not copied to OpenWrt; each concept is translated to its OpenWrt equivalent (Table 9). The equivalents come from the OpenWrt documentation (OpenWrt Project, n.d.-a, n.d.-b, n.d.-c, n.d.-d, n.d.-e); each one will be validated on the hardware in weeks 10 and 11.

Table 9

Translation of Cisco Concepts to OpenWrt

Cisco concept	Prototype status	OpenWrt equivalent	How it will be validated
VLAN database, access ports	PENDING	Bridge VLAN filtering on DSA ports (*bridge-vlan* sections in the network configuration) (OpenWrt Project, n.d.-b, n.d.-e)	A PC on each VLAN gets the right subnet
802.1Q trunk, Router-on-a-Stick	PENDING	Tagged VLAN interfaces on the router port (eth0.N) (OpenWrt Project, n.d.-e)	Inter-VLAN ping through the router
EtherChannel (LACP) and RSTP	PENDING	Link bonding and bridge spanning tree on the Linux side; the Cisco switches keep their own LACP and RSTP. Details to be confirmed in week 10	Unplug one member link; traffic continues
DHCP pools and excluded addresses	Implemented	dnsmasq: *dhcp* sections with start, limit, and static leases (OpenWrt Project, n.d.-a)	Same lease range as in Table 5
Static routes	Implemented (temporary)	Static route entries in the network configuration (OpenWrt Project, n.d.-d)	Same pings and traces as T02 to T05
OSPF	PENDING	FRRouting *ospfd*, area 0 on the shared segment (FRRouting, n.d.)	Neighbor state Full; routes learned
NAT and firewall	PENDING	fw4 (nftables) zones, masquerading excluding 10.10.0.0/16 (OpenWrt Project, n.d.-c)	Internet access per site; inter-site traffic not translated
Hashed passwords, SSH	Implemented in the simulation	Dropbear SSH with key or hashed credentials; no plain text in the repository	Same wrong-then-correct login tests as T12 to T14

3.8 Updated Schedule Through Week 11

Table 10 shows the plan. The roles are those of the cover page, and all three members take part in every stage.

Table 10

Schedule From Week 9 to Week 11

Week	Work	Role	Deliverable
9 (this delivery)	Prototype of the user LANs, DHCP, static routes, hashed logins and router SSH, tests T01–T14, report	All	PDF, .pkt, configurations, evidence
10	VLANs 10/20/30/99, trunk, Router-on-a-Stick, guest and server subnets; test P01–P03	Student 1	Updated .pkt and addressing in use
10	Stronger unique secrets, VTY access list, SSH to the switches once VLAN 99 exists; DHCP for guests	Student 3	Secure configuration (no secrets in files); tests P05, P10
10	Second switch per site, EtherChannel, RSTP; reverse trace test P08; test matrix update	Students 1 and 2	Failover evidence (P04)
11	OSPF replaces the static routes; NAT and firewall; full simulation run	Students 1, 2, and 3	Complete simulation (P06, P07)
11	Initial deployment on hardware: OpenWrt base, DHCP, routes, one VLAN	All	Hardware evidence (P09)

3.9 Team Contributions and Responsibilities

The three members of the team, Luis Angel Novillo Cruz, Arthur Brayan Beltran Hernández, and Alex Fernando Alban Torres, share the responsibility for every part of this delivery: the network design and addressing, the configuration and testing of the prototype, the secure administration of the devices, the evidence, the report, and the completion and migration plan. The areas on the cover page (network design, integration and validation, and security and services) show who leads each area, but the team agreed that all three members know the whole project and are able to explain and defend any part of it.

Pending tasks are listed in Table 7, the schedule through week 11 is in Table 10, and the plan to move the design to OpenWrt is in Table 9.

4. Conclusions and Future Work

- The two user LANs and the shared segment work in Packet Tracer: the four PCs obtained their addresses by DHCP and communicate inside each site and between the sites (T01–T06).
- All five devices require a login with hashed secrets, and both routers accept SSH (T12–T14); the switches still need a management address for SSH.
- The static routes give a two-router path, confirmed by the traceroute (T05); they are temporary and will be replaced by OSPF.
- The interface, routing, DHCP, and MAC table outputs agree with each other (T07–T11).
- VLANs, trunk, EtherChannel, RSTP, NAT, firewall, SSH to the switches, and the hardware tests remain PENDING.

Future work: implement the VLANs and the trunk in week 10, add the second switch, replace the current secrets with stronger ones, and complete OSPF, NAT, firewall, and the first hardware deployment in week 11.

5. Acknowledgments

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